

**Hey look we have a cluster now
isn't that cool what does it do does it do stuff yes it
does it's cool**

Doing cluster stuff for fun and ~~profit~~ science

Lukas Burk

Leibniz Institute for Prevention Research and Epidemiology — BIPS

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BioWiMium

We have a new cluster

You might have *heard*



1

What we cover today:

- What is this cluster thing
- How does it work (using R / Python, Slurm)
- How do **you** do stuff on it in general

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- Everything you ever need to know

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I know it's noisy and no I don't know when that is fixed

What is a cluster?

And why should you care



2

Many ways to shave a yak, and now you have one more!

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- You know it
- Shorter tasks
- Not so powerful
- Convenience: 🖱️

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- Bertha / new GePaRD server
- Run looong tasks
- 1 powerful computer
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HPC Cluster

- Many computers connected
- Access via *head node*
- Shared (scheduling: *Slurm*)

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What does it look like?

Kind of like this



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The HPC workflow

It's a freight truck, not a bicycle



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You don't hop on the cluster for a quick ride, you **submit work** to it

The HPC workflow

It's a freight truck, not a bicycle

You don't hop on the cluster for a quick ride, you **submit work** to it

What you are used to

- Open RStudio / VSCode / Positron
- Write code, run it, see results
- Everything is *interactive*
- One computer does it all

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- Prepare your code on the *head node*
- **Submit** it as a job (it gets *queued*)
- Go make coffee ☕
- Come back, check results
- Debug if needed, resubmit

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💡 There's a learning curve, but the payoff is huge

HPC Workflow

The typical cycle



5

1. Log in (land on **head node**)

HPC Workflow

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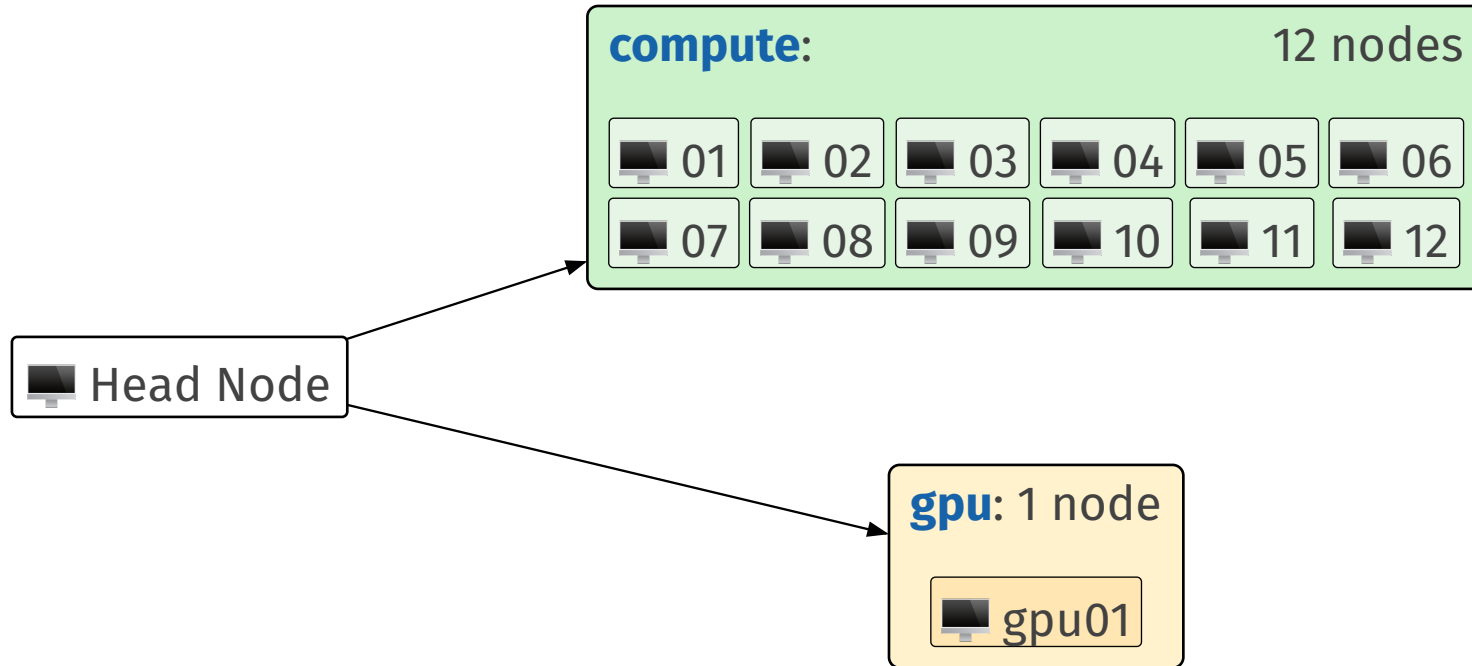


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8. Repeat until **insane** done

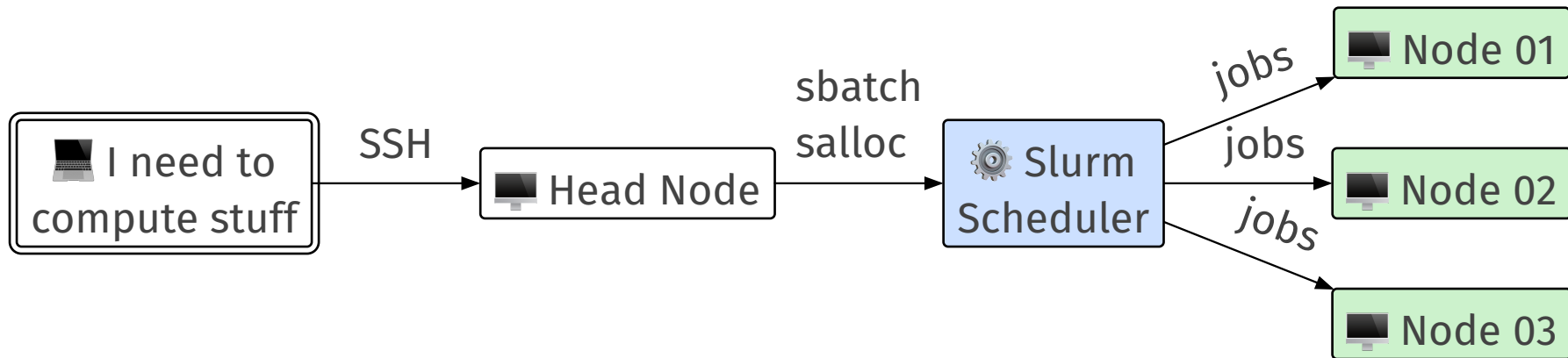
Cluster topology

Partitions



Cluster topology

It only looks weird because it is



The head node

Your gateway



8

The head node is where you land when you log in

The head node

Your gateway



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Do ✓

- Install R / Python packages
- Edit scripts and code
- Submit and monitor jobs
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Don't ✗

- Run heavy computations
- Launch long-running scripts
- Load huge datasets into memory

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⚠ The head node is shared by *everyone*. Heavy processes will be *terminated*.

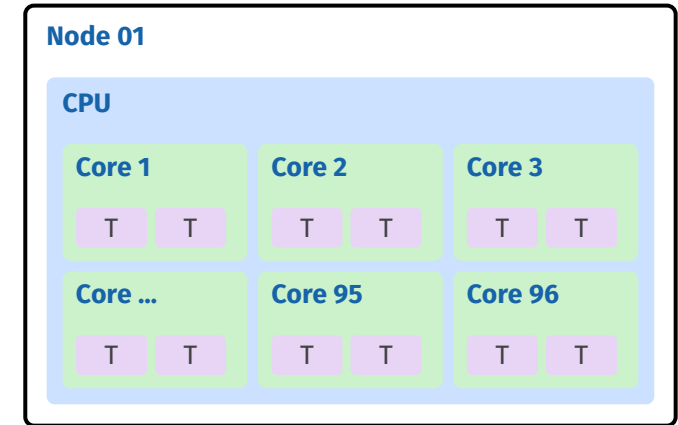
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Just to make sure you're confused



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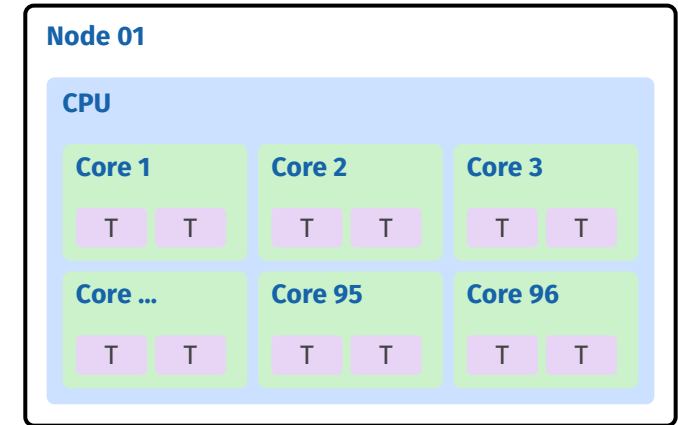
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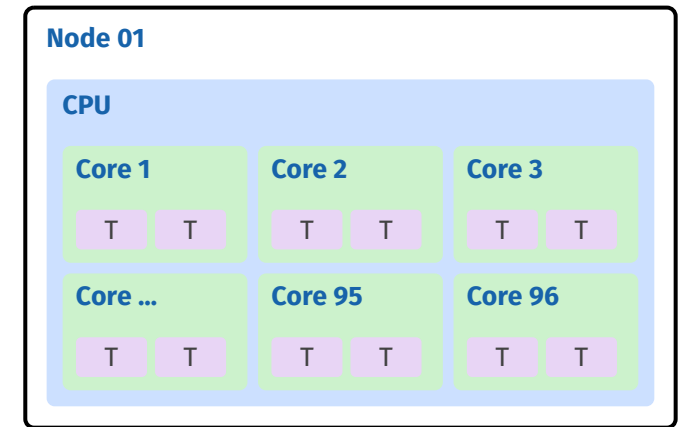
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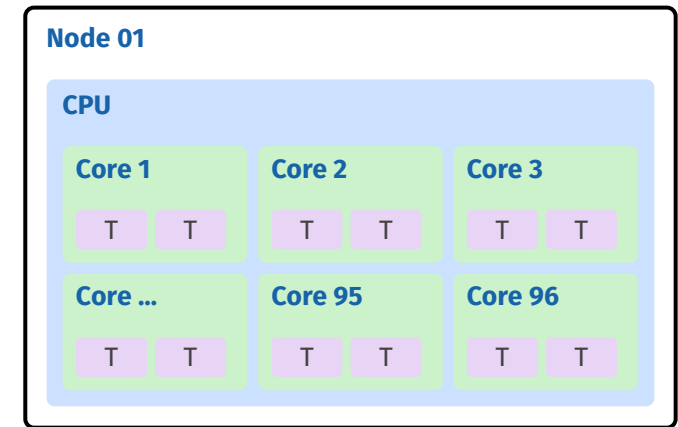
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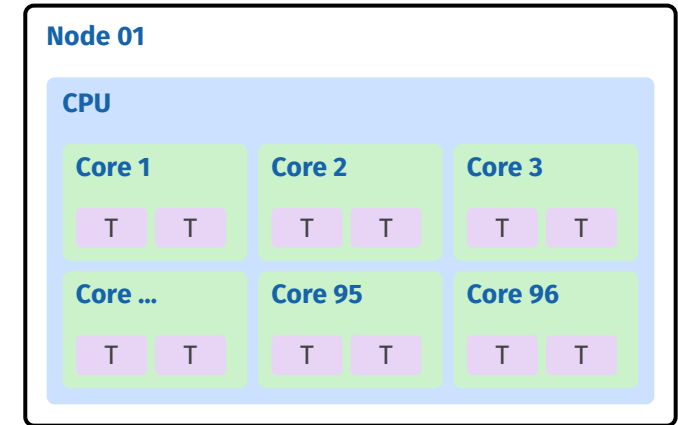
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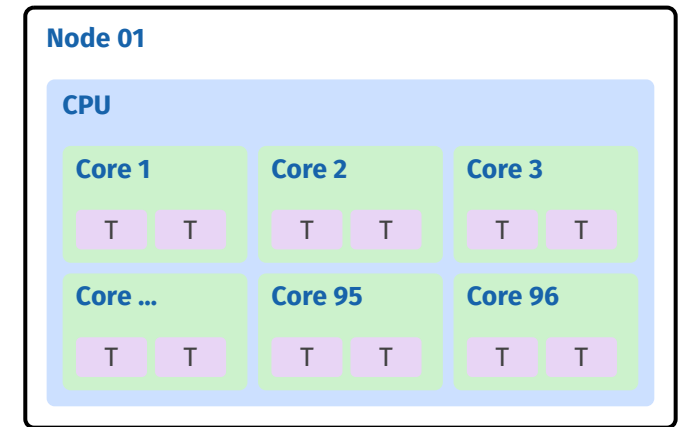
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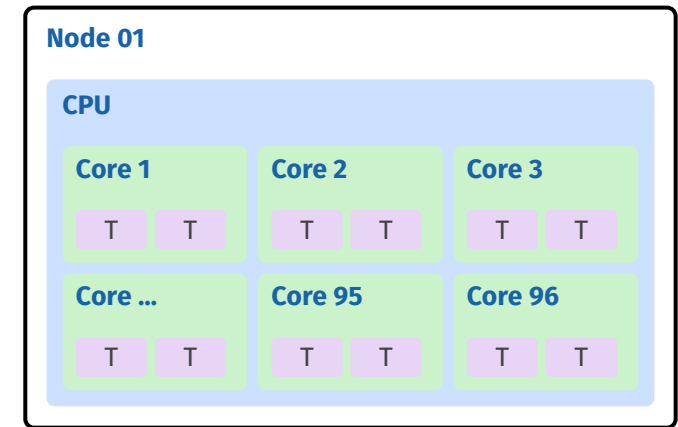
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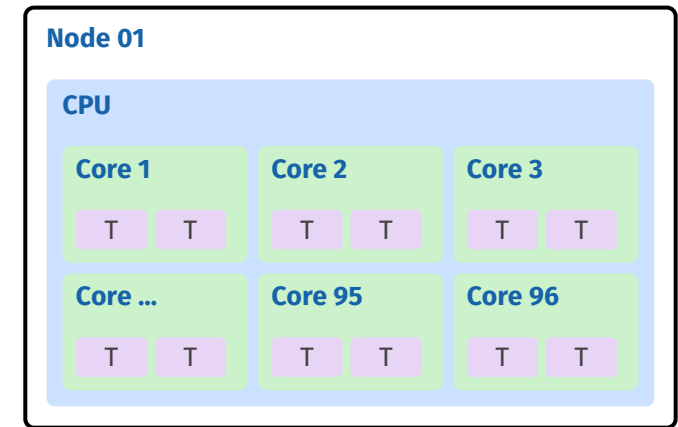
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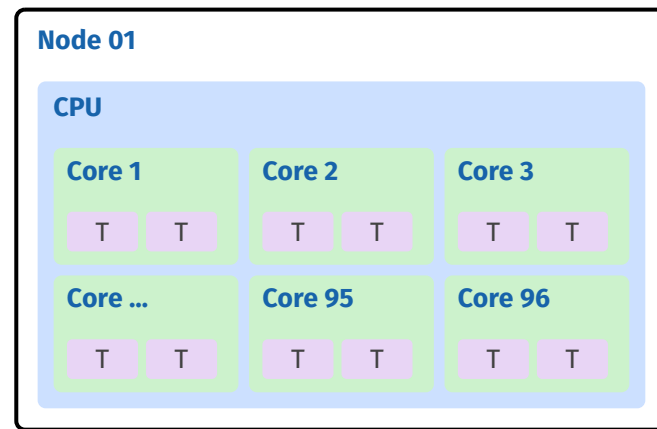
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⚠ **Slurm**: You need 10 **threads** → you request 10 “cpus”

But what is Slurm even

- Slurm is the job management system (see slurm.schedmd.com)
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→ It’s impossible to “endanger” other people’s compute jobs

Logging in

Different environments for different tasks



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Connect to the cluster **head node** via SSH in one of two ways:

Logging in

Different environments for different tasks

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Connect to the cluster **head node** via SSH in one of two ways:

Terminal

- Definitely cool and normal
- 1337 h4xX0r feeling

```
burk hnode ~/projects
```

Permissions	User	Group	Size	Date Modified	Name
wxr-xr-x	burk	burk	4.0K	2026-04-02 19:35	📁 bips-cluster-demos/
wxr-xr-x	burk	burk	4.0K	2026-03-19 00:12	📁 paper_2023_survival_benchmark/
wxr-xr-x	burk	burk	4.0K	2026-03-05 00:12	📁 paper_2025_xplainfi_benchmark/
wxr-xr-x	burk	burk	4.0K	2026-03-31 13:23	📁 reduction-techniques-benchmark/
wxr-xr-x	burk	burk	4.0K	2026-03-31 15:50	📁 reduction-techniques-benchmark-update/
wxr-xr-x	burk	burk	4.0K	2026-03-17 20:13	📁 resolve-pcc-symptoms-multi-analyst/
wxr-xr-x	burk	burk	4.0K	2026-03-18 01:19	📁 resolve-pcc-xai/
wxr-xr-x	burk	burk	4.0K	2026-04-01 18:31	📁 survival-benchmark/
wxr-xr-x	burk	burk	4.0K	2026-03-20 21:15	📁 xplainfi-benchmark/

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ls -la
drwxr-xr-x burk burk 4.0K 2026-04-02 19:35 bips-cluster-demos/
drwxr-xr-x burk burk 4.0K 2026-03-19 00:12 paper_2023_survival_benchmark/
drwxr-xr-x burk burk 4.0K 2026-03-05 00:12 paper_2025_xplainfi_benchmark/
drwxr-xr-x burk burk 4.0K 2026-03-31 13:23 reduction-techniques-benchmark/
drwxr-xr-x burk burk 4.0K 2026-03-31 15:50 reduction-techniques-benchmark-update/
drwxr-xr-x burk burk 4.0K 2026-03-17 20:13 resolve-pcc-symptoms-multi-analyst/
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```

Positron (VScode)

- Uses SSH for communication with head node
- GUI on your own device (PC, laptop)
- Terminal / R console runs **on head node**
- All files you see are on the head node
- 😎 Edit files conveniently

The first thing you see when you log in

Trying to point you in the right direction



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A large ASCII art representation of a coffee cup, rendered in a light blue color on a dark background. The cup is filled with a grid-like pattern, and there's a small handle on the right side.

Welcome to the BIPS HPC cluster.

Dashboard: <http://srvcluster.bips.de>
Docs: <https://cluster.bips.coffee>
Support: Contact Lukas I guess?

This is the HEAD NODE -- for login, package installation, and job submission.

Do NOT run computations here.

- Use **salloc** for interactive sessions or
- Use **sbatch** for batch jobs

Run **slurm_user_help** to see available Slurm command aliases

If you don't know what any of this means, start with the docs linked above

Software is managed differently

Environment modules

- Common on HPC systems: Hundreds of users need dozens of programs
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Solution: Environment modules

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- Run `module load R/4.5.3` → R v4.5.3 is available
- Load modules **before** running `salloc` or `sbatch` — they inherit your environment

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- (Python: Use `uv`, no need for modules: (docs.astral.sh/uv))

How are resources allocated?

Slurm manages resources on the cluster



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Interactive: salloc

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salloc --cpus-per-task=20 --mem=4G --time=06:00:00
```

```
salloc: Granted job allocation 137031
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```
salloc: Nodes node01 are ready for job
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# start R session
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R
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# Do work or run script
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ranger::ranger(foo ~ ., data = bar)
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Batch job: sbatch

- Write an analysis.sh wrapper script

```
#!/bin/bash
```

```
#SBATCH --job-name=my-analysis
```

```
#SBATCH --output=logs/%x_%j.out
```

```
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```
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```

```
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```
Rscript analysis.R
```

- Run sbatch analysis.sh and go home

Slurm resource allocation

Partitions and QoS



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- Slurm groups hardware in *partitions*
- 2 partitions: *compute* (12 nodes, *default*) and *gpu* (1 node)
- Prioritization is done in *Quality of Service* (QoS) queues:

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- For `sbatch` or `#SBATCH` `--qos=<name>`:

short: 1 hour

medium (default): 1 day

long: 7 days

Slurm resource allocation

A job is not a job

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- → More efficient to just ask for 2 “cpus” and 1 full core

Example: Efficiently using a node

If your goal is to fully utilize 1 node (96 cores, 192 threads)

What works

- 1 job with 192 threads ✓

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If your goal is to fully utilize 1 node (96 cores, 192 threads)

What works

- 1 job with 192 threads ✓
- 96 jobs with 2 threads ✓

Example: Efficiently using a node

If your goal is to fully utilize 1 node (96 cores, 192 threads)

What works

- 1 job with 192 threads ✓
- 96 jobs with 2 threads ✓
- 8 jobs with 24 threads ✓

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- Each job reserves 1 **core**
- → 192 jobs allocate 192 cores, but a node only has 96!

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⚠ If you're used to `batchtools`, you probably default to single-threaded jobs

Did it work?

Checking on your jobs



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While running:

- `queue --me`: list your jobs
- `tail -f logs/job_12345.out`: watch output
- `scancel <jobid>`: cancel a job

After completion:

- Check your output files (e.g. `logs/`)
- `sacct --starttime=today`: job history
- `sacct -j <jobid> --format=JobID,State,Elapsed,MaxRSS`

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Common failure modes

- **TIMEOUT**: job ran out of time, request more with `--time`
- **OUT_OF_MEMORY**: request more RAM with `--mem`
- **FAILED**: your script has a bug, check the log output (*did you load modules?*)

Where does your stuff live?

Two storage tiers



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Home directory

`/srv/home/<user>`

- Fast
- Limited space, also shared with software
- Scripts, code, *active* projects

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Two storage tiers

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Archive storage

/mnt/sas/users/<user>

- Large capacity
- Slower
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- Both are **shared across all nodes** (head + compute) via internal network
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⚠ There are **no backups**. Use `git` for code, be careful with data.

User utilities

Command-line aliases/functions



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- I have prepared a few shorthand tools based on things I need often
- On head node, get info with `slurm_user_help`:

```
> slurm_user_help
=== Slurm User Helper Functions ===
```

JOB MONITORING:

```
sq          - Enhanced squeue with better formatting
sqm         - My jobs queue
sqs         - Job status summary with colors
[...]
```

JOB ACCOUNTING:

```
slac        - Enhanced sacct
slac1h/1d/3d/1w - Jobs from last hour/day/3days/week (excludes PENDING)
  State filters: -a (all), -p (pending), -c (completed), -f (failed), -r (running)
sltoday     - Jobs submitted today
```

Getting started

And further reading material



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1. Ask me for an account, I will send you login credentials
2. Read (and bookmark) the docs:
 - cluster.bips.coffee (public)
3. Bookmark the dashboard (current cluster usage):
 - <http://srvcluster.bips.de/> (no https!)
4. Try demos / usage examples for batchtools, mirai, targets:
 - srvgit.bips.eu/bips/bips-cluster-demos